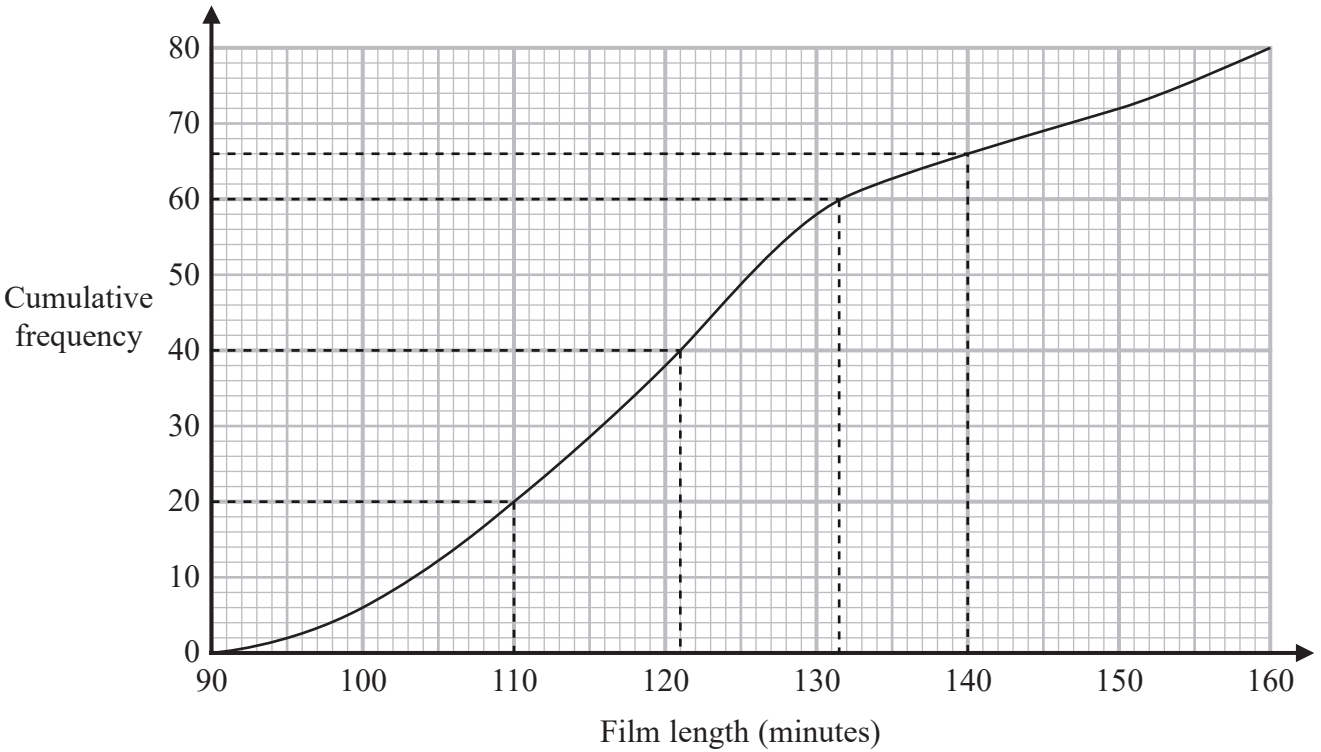
8 The cumulative frequency graph shows information about the lengths, in minutes, of 80 films.



(a) Use the graph to find an estimate for the median film length.

Median is at 80 ÷ 2 = 40 films.

121 minutes

1 mark

Read across at cumulative frequency 40, then down.

(1)

(b) Use the graph to find an estimate for the interquartile range of the film lengths.

Lower quartile at 20 films: 110 minutes

Upper quartile at 60 films: about 131.5 minutes

1 mark

Read the quarter and three-quarter positions.

..... minutes

IQR ≈ 131.5 – 110 = 21.5 minutes

(2)

1 mark

Subtract the lower quartile from the upper quartile.

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(c) Use the graph to find an estimate for the percentage of these films that are more than 140 minutes long.

At 140 minutes, cumulative frequency ≈ 66.

1 mark
Read the number at or below 140 minutes.

Number above 140 minutes ≈ 80 – 66 = 14

1 mark
Subtract from 80, then divide by the total and multiply by 100.

Percentage ≈ 14 ÷ 80 × 100 = 17.5%

17.5 %

1 mark
Use the number above 140.

(3)

(Total for Question 8 is 6 marks)